Troubleshooting Branch-to-Branch IPsec VPN Connectivity

1. Objective

To troubleshoot a site-to-site IPsec VPN scenario in which devices at one branch office can access the Internet but cannot access devices at the other branch. The objective is to identify the VPN traffic-selection problem, document the troubleshooting commands, correct the configuration, and verify branch-to-branch connectivity.

2. Scenario and Topology

```
Branch A                                     Branch B
PC0/PC1/PC2 — Switch — Router1 ===== WAN ===== Router0 — Switch — PC3
192.168.1.0/24         200.1.1.2   200.1.1.1          192.168.2.0/24
```

Router1 represents Branch A and Router0 represents Branch B. The serial WAN connection between the routers is operational. Internet connectivity from Branch A is working, but traffic destined for the Branch B LAN initially fails through the VPN.

3. IP Addressing

Device	Interface	IP Address	Role
Router1	G0/0/0	192.168.1.1/24	Branch A gateway
Router1	S0/1/0	200.1.1.2/30	WAN / VPN peer
Router0	S0/1/0	200.1.1.1/30	WAN / VPN peer
Router0	G0/0/0	192.168.2.1/24	Branch B gateway
PC0	Fa0	192.168.1.10/24	Branch A host
PC1	Fa0	192.168.1.11/24	Branch A host
PC2	Fa0	192.168.1.12/24	Branch A host
PC3	Fa0	192.168.2.10/24	Branch B host

4. Problem Statement

The branch network could reach the Internet, confirming that the basic LAN, WAN, routing, and Internet path were functioning. However, a host in Branch A could not reliably access the Branch B host at 192.168.2.10 through the intended IPsec VPN.

```
PC0> ping 8.8.8.8
PC0> ping 192.168.2.10
```

The Internet ping was successful while the branch-to-branch test was unsuccessful. This indicated that the fault was specific to the VPN path or the traffic selected for encryption.

5. Troubleshooting Commands

5.1 Check ISAKMP Status

```
Router1# show crypto isakmp sa
```

This command was used to check whether the ISAKMP security association was established. An established tunnel is commonly shown with a state such as QM_IDLE.

5.2 Check IPsec Security Associations

```
Router1# show crypto ipsec sa
```

The IPsec SA output was checked for packet counters, especially #pkts encaps and #pkts decaps. These counters help determine whether traffic is actually entering and leaving the IPsec tunnel.

5.3 Check Access Lists

```
Router1# show access-lists
```

The VPN crypto ACL was inspected because an IPsec crypto map uses an ACL to identify the interesting traffic that must be encrypted.

5.4 Check the Running Configuration

```
Router1# show run | section access-list
```

The running configuration was reviewed to compare the configured VPN traffic selectors with the actual Branch A and Branch B networks.

6. Fault Identified

The problem was an incorrect destination network in Router1's crypto ACL. The actual Branch B network was 192.168.2.0/24, but the ACL incorrectly referenced 192.168.3.0/24.

Incorrect configuration:

```
access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

Because 192.168.2.0/24 did not match the crypto ACL, the branch-to-branch traffic was not selected as IPsec interesting traffic.

7. Configuration Change / Solution

The incorrect ACL was removed and replaced with the correct Branch A-to-Branch B selector:

```
Router1# configure terminal
Router1(config)# no access-list 110
Router1(config)# access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router1(config)# end
```

The corrected ACL now matches:

```
192.168.1.0/24  →  192.168.2.0/24
```

8. Verification

After correcting the ACL, the branch-to-branch connectivity test was repeated:

```
PC0> ping 192.168.2.10
```

Successful replies confirmed that Branch A could communicate with the Branch B host.

The VPN was then verified with:

```
Router1# show crypto isakmp sa
Router1# show crypto ipsec sa
```

The ISAKMP state should be established (commonly QM_IDLE), and the IPsec encapsulation and decapsulation counters should increase when branch-to-branch traffic is generated.

Result

The connectivity problem was successfully diagnosed and resolved. Internet access remained available, while the branch-to-branch VPN traffic initially failed because Router1's crypto ACL specified the wrong destination network. Updating the ACL to permit traffic from 192.168.1.0/24 to 192.168.2.0/24 restored communication between the two branch offices.